INCH-POUND

MIL-PRF-1/1614B
9 July 1998
SUPERSEDING
MIL-E-1/1614A
12 December 1974

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, POWER
TYPE 857B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Half-wave mercury rectifier.

See figure 1.

Mounting position: Base down.

Weight: 3.5 pounds (1.59 kilograms).

ABSOLUTE RATINGS:

Parameter:	Ef	Epx	ib	Io	T Hg	tk	Frequency
Unit:	Vac 1/	kv	a	A dc	°C	sec	Hz
Maximum	5.25	22	40	10	40	----	150
Minimum	4.75	----	----	----	30	60	25
Maximum	5.25	10	40	10	60	----	150
Minimum	4.75	----	----	----	25	60	25
Test 4/ Condition:	5.0	25	----	----	30 to 40	60	60

1/ Increased life may be obtained by using filament 90° ± 30° out of phase with anode excitation.

GENERAL:

Qualification - Required.

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TABLE I. Testing and inspection.

Inspection	Method	Conditions	Acceptance level	Inspection level or code	Symbol	Limits		Unit
						Min	Max	
Conformance inspection, part 1								
Peak emission by voltage drop	1231	Ib = 40 a	0.65	II	etd	----	20	v
Filament current	1301		0.65	II	If	27	33	Aac
Operation of rectifiers 1/ 2/	1353	I_o = 10 A dc (min); t = 120	0.65	II	----	----	----	----
Conformance inspection, part 3								
Service - life guarantee	----	t = 1,000 hours	----	----	----	----	----	----

1/ Artificial control of condensed mercury temperature may be used for this test, half specified inverse voltage may be applied initially and then gradually increased to full voltage in approximately 1 minute.

2/ This test is to be the first test performed at the conclusion of the holding period.

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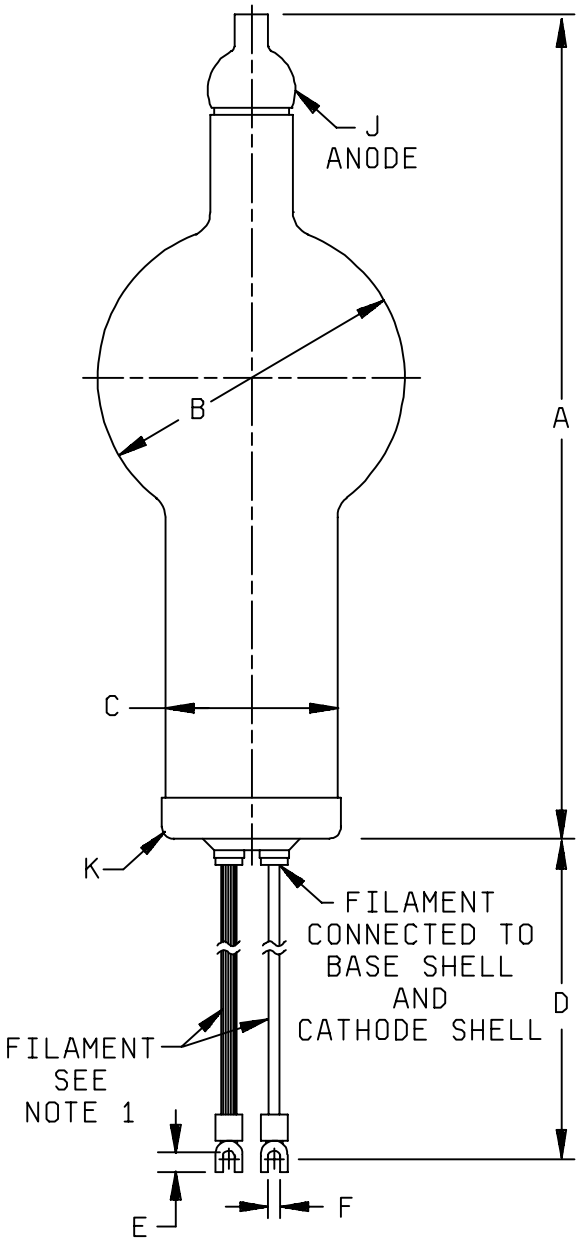

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
Conformance inspection, part 2				
A	19.120	19.880	485.65	504.95
B	----	7.130	----	181.10
Conformance inspection, part 3 (see note b)				
C	----	4.130	----	104.90
D	7.500	10.000	190.50	254.00
F	0.310	0.380	7.87	9.65
J	Cap: C1 - 10 (EIA)			
K	Base: F0 - 2 or F0 - 1 (EIA)			
Reference dimension				
E	0.625		15.88	

NOTES:
1. Material used for insulated sleeve may be beaded ceramic or vinyl coated fiberglass or equivalent.

2. A sample size of three tubes shall be checked annually, with no failures allowed. In case of a sample failure, the test shall become a conformance inspection, part 2 test, for three successful consecutive submissions at which time the test may revert to the conformance inspection, part 3 test.

FIGURE 1. Outline drawing of electron tube type 857B .

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - AS, CG, MC, OS, SH
Air Force - 11, 17, 99

Preparing activity:
DLA - CC

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